1. Clinical Question & Core Hypothesis

Clinical Question

TLS-rich (or lymphoid-aggregate–rich) tumors are often thought to indicate a pre-existing anti-tumor immune response and thus predict better response to immune checkpoint blockade, but in HCC this association is inconsistent — why do some TLS-rich tumors fail to respond to PD-1/PD-L1 blockade? 2) Tier 1 paths used: the direct S100A9 → PD-L1 path provides a mechanistic backbone: inflammation-driven S100A9 can induce or modulate PD-L1, offering a plausible molecular switch that converts an apparently immune-active microenvironment into an immunosuppressive one. I built on this because it links an inflammatory mediator to checkpoint expression, directly addressing the TLS→therapy_response puzzle. 3) Tier 2 evidence supporting the claim: EPB41L2, Osteopontin (SPP1), and FGFR3 appear downstream of risk-factor and prognostic axes and could modulate the inflammatory/risk milieu (e.g., myeloid recruitment, matrix remodeling, growth-factor signaling) that enables S100A9-driven PD-L1 upregulation or affects TLS functionality; these therefore serve as secondary modifiers to test the broader context. 4) Why therapy_response was chosen as the endpoint: PD-L1 is central to PD-1/PD-L1 immune checkpoint blockade, so the most direct clinical consequence of S100A9→PD-L1 signaling is altered response to ICI; this is a testable, clinically actionable endpoint. 5) Deliberate exclusions: I excluded the broad upstream 'signaling network' and the full list of risk-factor descriptors from the core hypothesis to keep the mechanistic claim sharp and testable; EPB41L2, SPP1 (Osteopontin), and FGFR3 were moved to secondary_modifiers rather than included in the core claim to avoid overfitting the primary mechanistic statement to many variables?

Working Hypothesis

In TLS-rich hepatocellular carcinoma (HCC), high S100A9 expression drives PD-L1 upregulation on myeloid/antigen-presenting cells within the TLS microenvironment through TLR4/RAGE-mediated signaling, activating NF-κB and promoting an immunosuppressive niche that blunts CD8+ T cell function. Consequently, TLS-rich, S100A9-high and PD-L1-high tumors exhibit reduced response to PD-1/PD-L1 blockade compared with TLS-rich tumors lacking S100A9-driven PD-L1 induction. Secondary modifiers EPB41L2, SPP1, and FGFR3 modulate the magnitude or spatial pattern of this axis by influencing myeloid recruitment, extracellular matrix remodeling, and growth-factor signaling, thereby creating heterogeneity in therapeutic response within TLS-rich HCCs. Disruption of S100A9 signaling (genetic or pharmacologic) and/or modulation of the downstream modifiers will attenuate PD-L1 upregulation, restore T cell effector function, and improve responsiveness to checkpoint blockade in preclinical models. This yields testable predictions: (i) S100A9-high/TLS-rich tumors will have higher PD-L1 and exhausted T-cell phenotypes and poorer responses to anti-PD-1/PD-L1; (ii) inhibition of S100A9 or TLR4/RAGE axis will reduce PD-L1 induction and enhance therapy efficacy; (iii) manipulating EPB41L2, SPP1, or FGFR3 will alter the S100A9–PD-L1 axis and TLS functionality, modifying treatment outcomes.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	S100A9, PD-L1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

S100A9: S100A9 → PD-L1 path provides a mechanistic backbone: inflammation-driven S100A9 can induce or modulate PD-L1, offering a plausible molecular switch that converts an apparently immune-active microenvironment into an immunosuppressive one.

PD-L1: PD-L1 blockade? 2) Tier 1 paths used: the direct S100A9 → PD-L1 path provides a mechanistic backbone: inflammation-driven S100A9 can induce or modulate PD-L1, offering a plausible molecular switch that converts an apparently immune-active microenvironment into an immunosuppressive one.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	EPB41L2	Supporting modifier from knowledge graph
Secondary	SPP1	Supporting modifier from knowledge graph
Secondary	FGFR3	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: manipulating S100A9 and measuring PD-L1 and NF-κB signaling;

Prediction 2: testing PD-1/PD-L1 blockade in TLS-rich, S100A9-high contexts with and without S100A9 axis inhibition;

Prediction 3: evaluating secondary modifiers' impact on the axis and on TLS architecture;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- EPB41L2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- SPP1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.